NOT MEASUREMENT SENSITIVE

MIL-T-62491C
26 February 1992
SUPERSEDING
MIL-T-62491B
26 October 1989

MILITARY SPECIFICATION

TRUCKS, WRECKER: 4540 TO 40 820 KILOGRAMS (5-TON TO 45-TON) LIFTING CAPACITY, MODIFIED COMMERCIAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers all commercial, non-tactical wreckers procured for military use. Vehicles procured under this specification are commercial items which are warranted by the manufacturer as specified in acquisition documents.

1.2 Vehicle classification. The vehicle shall be one of the types, classes, sizes and models shown in table I, as specified (see 6.2):

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-UED, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC-2320

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TABLE I. Vehicle classification.

Type	Class	Size	Model
I – Hydraulic	10	5T	4x2 or 4x4
	10	7T	4x2 or 4x4
	28	20T	4x2 or 4x4
	32	25T	4x2 or 4x4
	39.5	35T	6x4 or 6x6
	39.5	40T	6x4 or 6x6
	50	45T	6x4 or 6x6
II - Mechanical	21	16T	4x2
	24	24T	4x2 or 4x4
	32	24T	4x2 or 4x4
	34.5	24T	6x4
	36	24T	6x6
	44.5	24T	6x4 or 6x6

1.2.1 Class. The class of the vehicle shall be determined by the minimum GVW rating, as shown in table II:

Table II. Vehicle classes.

Class	GVW kilograms (kg) (pounds)
10	4550 (10,000)
21	9530 (21,000)
24	10 900 (24,000)
28	12 700 (28,000)
32	14 500 (32,000)
34.5	15 650 (34,500)
36	16 350 (36,000)
39.5	17 900 (39,500)
44.5	20 200 (44,500)
50	22 700 (50,000)

1.2.2 Size. The size of the vehicle shall be determined by the minimum lifting capacity of the wrecker boom(s), as shown in table III:

Table III. Vehicle sizes.

Size	Lifting capacity kg (pounds)
5T	4540 (10,000)
7T	6350 (14,000)

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Table III. Vehicle sizes. – (continued)

Size	Lifting capacity kg (pounds)
16T	14 500 (32,000)
20T	18 150 (40,000)
24T	21 750 (48,000)
25T	22 700 (50,000)
35T	31 800 (70,000)
40T	36 300 (80,000)
45T	40 800 (90,000)

1.2.3 Mode1. The model of the vehicle shall be determined by the number of wheels and the number of wheels driving, as shown in table IV:

Table IV. Vehicle models.

Model	Wheels	Wheels driving
4x2	4	2
4x4	4	4
6x4	6	4
6x6	6	6

2. APPLI CABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS), and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131 - Battery, Storage: Vehicular, Ignition, Lighting and Starting.

VV-F-800 - Fuel Oil, Diesel.

MILITARY

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.

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MIL-M-43719	- Marking Materials and Markers, Adhesive, Elastometric, Pigmented, General Specification for.
MIL-B-46176	- Brake Fluid, Silicone, Automotive All Weather, Operational and Preservative.
MIL-T-83133	- Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-H-83282	- Hydraulic Fluid, Fire-Resistant, Synthetic Hydrocarbon Base, Aircraft, NATO Code Number H-537.

STANDARDS

FEDERAL

FED-STD-297	- Rustproofing of Commercial (Non-tactical) Vehicles.

MILITARY

MIL-STD-1223	- Non-tactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plate Standards.
MS 51118	- Pintle Assembly, Towing - 40,000 Lbs. Capacity, Manual Release.
MS 51317	- Light, Warning, Vehicular, Rotating, D.C.
MS 51335	- Pintle Assembly, Towing - 18,000 Lbs. Capacity, Manual Release.
MS 75020	- Connector, Plug, Electrical - 12 Contact, Intervehicular, 28-Volt, Waterproof.
MS 75021	- Connector, Receptacle, Electrical – 12 Contact, Intervehicular, 28-Volt, Waterproof.

* (Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF DEFENSE (DOD)

Department of Defense Index of Specifications and Standards (DODISS).

(Copies of the DODISS are available on a yearly subscription basis either from the Government Printing Office for hard copy, or microfiche copies are available from the Director, Navy Publication and Printing Service Office, 700 Robbins Avenue, Philadelphia, PA 19111-5093.)

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DEPARTMENT OF TRANSPORTATION (DoT)
Federal Motor Carrier Safety Regulations.
Federal Motor Vehicle Safety Standards.

(Application for copies of DoT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)
Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.
Noise Emission Standards for Transportation Equipment - Medium and Heavy Trucks.

(Application for copies of EPA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 53	- Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless for Ordinary Use (DoD Adopted).
ASTM B 117	- Salt Spray (Fog) Testing Methods (DoD Adopted).
ASTM D 4985	- Standard Specification for Low Silicate Ethylene Glycol Base Engine Coolant for Heavy Duty Engines Requiring an Initial Charge of Supplemental Coolant Additive.

(Applications for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

THE EUROPEAN TYRE AND RIM TECHNICAL ORGANISATION (ETRTO)
Standards Manual

(Application for copies of the ETRTO publications should be addressed to the European Tyre and Rim Technical Organisation, 32, Avenue Brugmann, 1060 Brussels, Belgium.)

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SAE, INC.

SAE Standards and Recommended Practices.

	J318	- Air Brake Gladhand Service (Control) and, Emergency (Supply) Line Couplers –Truck, Truck-Tractors, and Trailers (DoD adopted).
*	J350	- Spark Arrester Test Procedure for Medium Size Engines (DoD adopted).
	J537	- Storage Batteries.
	J551	- Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices (30-1000 MHz).
	J560	- Seven-Conductor Electrical Connector for Truck-Trailer Jumper Cable.
	J588	- Turn Signal Lamps.
	J589	- Turn Signal Switch.
	J682	- Rear Wheel Splash and Stone Throw Protection (DoD adopted).
	J683	- Tire Chain Clearance - Trucks, Buses, and Combinations of Vehicles.
	J688	- Truck Ability Prediction Procedure.
	J704	- Openings for Six- and Eight-bolt Truck Transmission Mounted Power Take-Offs.
	J706	- Rating of Winches.
	J844	- Nonmetallic Air Brake System Tubing (DoD adopted).
	J1067	- Seven-Conductor Jacketed Cable for Truck-Trailer Connections.
	J1142	- Towability Design Criteria - Passenger Cars and Light Duty Trucks.
	J1349	- Engine Power Test Code - Spark Ignition and Diesel.

(Application for copies of SAE publications should be addressed to SAE, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

THE TIRE AND RIM ASSOCIATION, INC

Year Book.

* (Application for copies of The Tire and Rim Association publications should be addressed to the Tire and Rim Association, Inc., 175 Montrose West Ave., Copley, OH 44321.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard vehicle and accessories. Except as specified in 3.1.1 through 3.1.1.17, the vehicle, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items which meet or exceed the requirements of this specification. Except as specified in 3.1.1 through 3.1.1.17, no removal, substitution or alteration of the chassis manufacturer's standard or optional chassis components shall be made. All chassis items shall be as represented in the chassis manufacturer's technical data book. Special bodies or mounted equipment shall be as represented in the body and equipment manufacturer's technical data. Technical data shall be limited to specifications and technical material identical to that furnished to the authorized company representatives for selection of vehicle models and components, and shall be available to the engineering office of the procuring activity prior to delivery of the items. The chassis model furnished shall be not older than the chassis manufacturer's current model on the date of invitation for bids.

3.1.1 Special requirements. In addition to the standard vehicle and components specified in 3.1, the vehicle shall be furnished with special equipment as specified herein.

3.1.1.1 Treatment and painting. The vehicle shall be treated and painted in accordance with MIL-STD-1223. As specified by the procuring activity for the appropriate military service (see 6.2), the exterior color shall be in accordance with MIL-STD-1223.

3.1.1.2 Markings and data plates. As specified by the procuring activity for the appropriate military service (see 6.2), identification markings and data plates shall be in accordance with MIL-STD-1223. When specified (see 6.2), concealed markings shall be furnished.

3.1.1.3 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. When specified (see 6.2), tropical rustproofing in accordance with FED-STD-297 shall be furnished.

3.1.1.4 Drain plugs. Drain plugs installed in manual transmissions and rear axles shall be of the permanent magnet type.

* 3.1.1.5 Towing devices. Towing devices consisting of two hooks, loops, eyes or pins or the manufacturer's single center mounted eye or pin shall be mounted on the front of the vehicle. In addition, towing devices shall be mounted on the rear of the vehicle. All towing devices shall be frame rail mounted or reinforced back to each frame rail.

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3.1.1.6 Wheel sp1ash and stone throw protection. Rubber mud flaps shall be provided to the rear of the rear wheels. Tilt cabs shall also have rubber mud flaps to the rear of the front wheels. A metal strip not less than 3.2 millimeters (mm) 0.125 inch thick and not less than 25 mm (one inch) wide, extending the entire width of the mud flap, shall be installed to prevent the bolt heads or bolt nuts from damaging the mud flap. As an alternate method of attaching the mud flaps, tabs or clips with minimum surface contact dimensions of 25 mm (1 inch) high by 32 mm (1.25 inch) wide by 2.4 mm (0.094 inch) thick shall be furnished at each bolt. Mud flap installations front and rear shall conform to the rear wheel splash and stone throw protection provisions of SAE J682.

* 3.1.1.7 Trailer towing package, class 21. A trailer towing package shall be furnished on class 21. The trailer towing package shall consist of a pintle, safety chain attachment devices, a lighting receptacle and associated reinforcements and wiring installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51335-1. The pintle shall be installed on the backplate (see 3.5.4 and 3.6.4) with reinforcements to transfer a vertical tongue load of not less than 815 kilograms (kg) (1,800 pounds) and a horizontal drawbar load of not less than 80 kilonewtons (kN) (18,000 pounds) directly to the chassis frame rails. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength at least equal to the GVW of the wrecker furnished. The attachment devices shall be capable of accommodating a standard grab hook [(56 mm (2-3/16 inches) wide, 18.7 mm (47/64 inch) thick and 12.3 mm (31/64-inch) throat width)] for a 9.6 mm (3/8-inch) chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color-coded as specified therein, shall be mounted in a readily accessible location near the pintle. The pintle height shall be in proportion to the frame height.

* 3.1.1.8 Trailer towing package, classes 24 through 50. A trailer towing package shall be furnished, except on hydraulic wreckers with a wheel lift (see 3.5.12), on classes 24 through 50. The trailer towing package shall consist of a pintle, safety chain attachment devices, a lighting receptacle, a trailer brake control system and associated reinforcements and wiring installed on the rear of the vehicle. The pintle shall be of the rotating type conforming to MS 51118. The pintle shall be installed on the backplate with reinforcements to transfer a vertical tongue load of not less than 1815 kg (4,000 pounds) and a horizontal drawbar load of not less than 178 kN (40,000 pounds) directly to the chassis frame rails. Two trailer safety chain attachment devices, one adjacent to each side of the pintle, shall be provided. Each attachment device shall provide an ultimate strength of not less than 178 kN (40,000 pounds). The attachment devices shall be capable of accommodating a standard grab hook [(116 mm (4-9/16 inches) wide, 30 mm (1-3/16 inches) thick, 19.8 mm (25/32-inch) throat width)] for a 16 mm (5/8-inch) chain. The lighting receptacle, conforming to SAE J560 with its conductors connected and color-coded as specified therein, shall be mounted in a readily accessible location near the pintle. The trailer

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brake control system shall conform to 3.4.11.3. The pintle height shall be in proportion to the frame height.

3.1.1.9 Trailer lighting table. When specified (see 6.2), a trailer lighting cable conforming to SAE J1067 shall be furnished. The cable shall be coiled and shall be not less than 2800 mm (110 inches) long when fully extended. Both ends of the cable shall be equipped with a round plug conforming to SAE J560. The plugs shall be equipped with a grip for withdrawing from the connector sockets. The cable shall be packaged and stowed in the stowage box or compartment (see 3.5.13(g), 3.6.12(h)).

3.1.1.10 Electric trailer brake controller. When specified (see 6.2), the vehicle shall be provided with an electric trailer brake controller. The controller shall be installed in the truck cab and wired through the SAE J560 lighting receptacle. The controller shall operate on the 12-volt electrical system of the vehicle and shall include load control, hand lever, and accessories.

3.1.1.11 Silicone brake fluid. When specified for class 10 vehicles (see 6.2), and if available as the manufacturer's standard or optional brake fluid, brake fluid conforming to MIL-B-46176 shall be provided in the hydraulic brake system. A tag shall be placed near the master cylinder stating "CAUTION: USE SILICONE BRAKE FLUID ONLY, MIL-B-46176."

3.1.1.12 Hydraulic fluid identification plate. A decal or plate shall be installed near the wrecker hydraulic system filler cap and shall identify the type of hydraulic fluid to be used. (See 3.1.1.13).

* 3.1.1.13 Decals and data plates. The hydraulic fluid identification information and the power takeoff caution notice shall be on a standard decal or plate from the supplier of that item or shall conform to type I or type II, class 1 of MIL-M-43719 or composition A (class 1 or 2) or composition C of MIL-P-514.

3.1.1.14 Brushguard. Model 4x4 and 6x6 vehicles shall be equipped with a radiator and headlamp brushguard. When the headlights are recessed into and protected by the front bumper, a headlamp brushguard is not required.

3.1.1.15 Brake lights. At least one pair of brake lights shall override the four-way emergency flasher or the two systems shall be independent of each other. Modifications to the manufacturer's standard product to accommodate this requirement shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard. If additional lights are added to the vehicle, the lights shall be selected from the chassis manufacturer's standard matching hardware.

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* 3.1.1.16 All wheel drive conversion. The chassis manufacturer's standard 4x2 or 6x4 truck chassis may be modified to provide all wheel drive conforming to the requirements specified herein if:

(a) The conversion axle manufacturer's engineering department specifically approves and certifies that all such modifications meet the design requirements and standards of the conversion axle manufacturer. Certification shall be based on both design analysis and proving ground test reports, which shall be made available to the engineering and quality assurance offices of the procuring activity.
(b) The chassis manufacturer's front axle before conversion has the same load rating as the conversion axle to be installed.
(c) Components used in the all-wheel drive conversion are of current production.
(d) Components used in the all-wheel drive conversion are approved for the conversion application by the component manufacturers.
(e) The converted vehicle is certified to conform to Federal Motor Vehicle Safety Standard No.121, by the intermediate or final manufacturer.
* (f) Replacement headlights, if required to be added, shall meet the height requirement of not less than 560 mm (22 inches) and not more than 1370 mm (54 inches), measured above the road surface, in conformance with Federal Motor Vehicle Safety Standard No.108. Replacement headlights shall be equivalent in mounting, protection, range and precision of adjustment to the chassis manufacturer's original standard headlights.
(g) Unused headlight cavities are covered in a neat workmanlike manner, treated and painted to match the chassis cab color with treatment and painting equivalent to the chassis cab manufacturer's process for the remainder of the chassis cab. Cavities and their covers shall be rustproofed in accordance with 3.1.1.3.
(h) Complete installation drawings are available to the acquisition activity.

* 3.1.1.17 Auxiliary generator unit. When specified (see 6.2), an auxiliary 120 volt, 60 Hertz (Hz), single phase, gasoline powered, 5 kilowatt (kw) alternating current (ac) generator unit shall be provided. The generator unit shall be located at the rear of the wrecker and placed such that the location will not interfere with the operation of the boom(s).

3.2 General design.

3.2.1 Federal Motor Vehicle Safety Standards. The vehicle and furnished accessories shall comply with all Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

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3.2.2 Air pollution control. The vehicle shall comply with EPA Regulations governing Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture.

* 3.2.3 Sound level. The cab interior sound level shall not exceed 84 db(A) when measured in accordance with Federal Motor Carrier Safety Regulation 393.94. The vehicle exterior sound level shall conform to Environmental Protection Agency Noise Emission Standards for Transportation Equipment, Medium and Heavy Trucks.

3.2.4 Curb weight. The curb weight shall include the weight of the chassis and cab, with all attachments, accessories, and equipment; the body; and a full complement of fuel, lubricants and coolant. The curb weight on the front axle shall be adequate to produce the wrecker lift-tow load as specified herein. A chassis exceeding the minimums specified herein (axle, suspension and tire capacities; engine size; cab type; hood projection; cab and hood materials; chassis source selection; and any other functional, weight-adding options) shall be furnished as needed to enable the wrecker to comply with the lift-tow load specified.

* 3.2.5 Gross vehicle weight and gross axle weight ratings. The gross vehicle weight (GVW) and the gross axle weight ratings (GAWR) of the axles, springs and tires shall be not less than that specified in table V for the corresponding vehicle class. In addition, the rear GAWR shall be at least equal to the load imposed on the rear axle, measured at the ground, with the wrecker lifting and towing the required lift-tow load capacity as specified in table X or XII, sub-item (b).

TABLE V. GVW and GAWR minimum ratings.

	Class	GVW Kg (pounds)	Front GAWR kg (pounds)	Rear GAWR kg (pounds)
*	10	4540 (10,000)	1360 (3,000)	3400 (7,500)
	21	9530 (21,000)	3180 (7,000)	7710 (17,000)
*	24	10 900 (24,000)	3180 (7,000)	9070 (20,000)
*	28	12 700 (28,000)	4540 (10,000)	9070 (20,000)
*	32 (type I)	14 500 (32,000)	4540 (10,000)	10 430 (23,000)
*	32 (type II)	14 500 (32,000)	5450 (12,000)	9300 (20,500)
	34.5	15 650 (34,500)	4540 (10,000)	13 380 (29,500)
	36	16 350 (36,000)	4540 (10,000)	13 380 (29,500)
	39.5	17 900 (39,500)	4990 (11,000)	16 240 (35,800)
	44.5	20 200 (44,500)	4990 (11,000)	16 240 (35,800)
	50	22 700 (50,000)	6000 (13,220)	19 960 (44,000)

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3.2.6 Ratings. Vehicle ratings shall be the manufacturer's published ratings. Component and vehicular ratings shall not be raised to meet the requirements of this specification. When published ratings are not available, verification of ratings must be submitted to the engineering office of the procuring activity.

3.2.7 Overall width. The overall width of the vehicles, exclusive of tires and safety related items such as mirrors, lights, reflectors and rubber body fenders, shall be not more than 2440 mm (96 inches). The width over the tires shall be:

(a) Not more than 2540 mm (100 inches) for single rear axles rated at 10430 kg (23,000 pounds) or less
(b) Not more than 2540 mm (100 inches) for tandem rear axles rated at 20900 kg (46,000 pounds) or less
(c) Not more than 2590 (102 inches) for tandem rear axles rated at over 20900 kg (46,000 pounds) and up to and including 26300 kg (58,000 pounds).

3.2.8 Accessibility. The design of the vehicle and optional equipment shall permit access for routine servicing and shall permit access for replacement and adjustment of component parts and accessories with minimal disturbance of other components and systems.

* 3.2.9 Prohibited materials. Except as permitted in 3.4.11, asbestos materials shall not be used in any form in any part of the vehicle. No item, part or assembly shall contain radioactive materials in which the specific activity is greater than 0.002 microcurie per gram or activity per item equals or exceeds 0.01 microcuries.

3.3 Performance requirements. Performance requirements for model 4x4 and model 6x6 vehicles shall be met with the front wheel drive disengaged except that vehicles equipped with interaxle compensating devices shall meet the performance requirements with the front wheel drive engaged.

3.3.1 Speeds and gradeability. Speed and gradeability requirements shall be met with the vehicle loaded to specified GVW.

3.3.1.1 High speed gradeability. The vehicle shall ascend continuous grades specified in table VI for the corresponding vehicle class at 80 kilometers per hours (km/h) (50 miles per hour (mph)). Gradeability requirements shall be met with the main transmission in direct drive, and when a multispeed axle or auxiliary transmission is furnished, with the axle and auxiliary transmission in high speed range. Gradeability shall be verified with calculations in accordance with SAE J688 (see 6.3).

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* TABLE VI. Gradeability.

Class	GVW kg (pounds)	Percent of grade
10	4540 (10,000)	5.0
21	9530 (21,000)	3.0
24	10 900 (24,000)	2.5
28	12 700 (28,000)	2.0
32	14 500 (32,000)	1.5
34.5	15 650 (34,500)	1.5
36	16 350 (36,000)	1.5
39.5	17 900 (39,500)	1.5
44.5	20 200 (44,500)	1.5
50	22 700 (50,000)	1.5

3.3.1.2 Low speed. Low speed for vehicles with a manual transmission shall be calculated with the engine operating at not less than 35 percent of recommended governed speed, and shall provide not more than the vehicle speed specified in table VII for the corresponding number of available forward speeds.

TABLE VII. Vehicle low speed requirements.

Number of forward speeds	Maximum low speed km/h (mph)
5	8.0 (5.0)
7 or 8	6.4 (4.0)
9 or 10	4.8 (3.0)
13 or more	4.0 (2.5)

3.3.1.3 Maximum geared speed. The maximum geared speed at engine governed speed shall be not less than 93 km/h (58 mph) for 4x2 and 6x4 model vehicles and not less than 84 km/h (52 mph) for 4x4 and 6x6 model vehicles. Conformance to geared speed specified shall be determined by calculating in accordance with the following formula:

$$\text{Maximum geared speed (km/h)} = \frac{\text{Governed speed (rpm) x 1.609}}{\text{Total gear reduction x tire factor (see 6.3).}}$$

3.3.2 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified GVW, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified GVW, within the stopping distance requirements of Federal Motor Carrier Safety Regulation 393.52.

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3.4 Chassis components.

3.4.1 Engine. The engine furnished for the specified vehicle class shall be the chassis manufacturer's standard or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

3.4.1.1 Diesel engine. Unless otherwise specified (see 3.4.1.2), the vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke or four-stroke cycle diesel engine. Engine net power used in performance prediction calculations shall be determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel conforming to VV-F-800. In addition, when specified (see 6.2), the engine shall operate satisfactorily on grade JP-4 and grade JP-5 fuel conforming to MIL-T-5624 under emergency, short duration conditions and on grade JP-8 fuel conforming to MIL-T-83133 under normal conditions. A power loss when operating on JP-4, JP-5 or JP-8 is acceptable.

* 3.4.1.2 Gasoline engine. When specified (see 6.2), the vehicle shall be equipped with a liquid cooled, internal combustion, four-stroke cycle gasoline engine. The engine furnished shall produce the required vehicle performance when operated on unleaded fuel with a research octane rating of 91, at an engine speed of not more than the manufacturer’s recommended operating speed. The engine shall be capable of warranted operation on unleaded fuel when used in accordance with the operator’s manual. Engine net power used in performance prediction calculations shall be determined in accordance with SAE J1349. Unless otherwise specified or required by foreign laws, vehicles for overseas use shall be capable of accepting and operating on leaded gasoline.

* 3.4.1.3 Oil filter. A full flow or combination full flow and bypass oil filter with replaceable element shall be furnished.

3.4.1.4 Governor. Except for class 10 wreckers, an engine governor shall be furnished, set and sealed to limit the engine to the engine manufacturer’s maximum recommended operating speed.

* 3.4.1.5 Cooling system. The cooling system shall maintain engine coolant at a temperature below the boiling point with the vehicle loaded and operated at an altitude of 3050 mm (10,000 feet) above sea level or in an ambient air temperature of not less than 52° Celsius (°C) (125 degrees Fahrenheit (°F)). The coolant system shall include a surge tank or a coolant recovery reservoir of not less than 1.89 liters (L) (two-quart) capacity. On tilt cab models, a radiator servicing access door shall be provided to allow ready verification of coolant level. For cooling system servicing, see 3.4.28.1.

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3.4.1.6 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. On diesel engine driven vehicles, the control shall include complete thermostatic control of all coolant flow through the radiator.

3.4.1.7 Fan clutch. A fan clutch shall be provided. The fan clutch shall reduce the fan speed automatically when the fan is not required for engine cooling. The fan clutch shall be asbestos free.

3.4.1.8 Silicone rubber hoses. When specified (see 6.2), except for class 10 wreckers, silicone rubber radiator and heater hoses shall be furnished.

* 3.4.1.9 Power plant heaters and fuel warmers. When specified (see 6.2), a coolant heater, an engine oil heater and fuel warmers (diesel engine driven vehicles only) shall be provided. Heaters shall operate on 110-volt alternating current (ac), and shall be wired through a junction block, including a fuse or circuit breaker, to a single three-pronged (male), weatherproof slave receptacle for receiving external power and grounding the vehicle. The receptacle shall be located on the streetside of the vehicle, as near the cab door as practicable. A three-wire connecting cable, not less than 7600 mm (25 feet) long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. Electrical apparatus shall conform to Federal Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperatures (down to -51°C (-60°F)) without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation, and without loose excess wire. Provisions for stowage of the cable shall be provided in the vehicle cab. Heaters and fuel warmers shall be furnished as follows:

(a) A coolant heater, 1000-watt (W) minimum rating, shall be installed in the engine block or lower coolant inlet hose. An engine thermostat with an operating range of 77°C (170°F) to 90°C (195°F) shall be installed.

* (b) An oil pan heater of the permanent external surface mount, permanent in-pan mount, or immersion type that meets the following requirements, shall be installed:

* (1) Not less than 16 W/L (15 watts per quart) heating capacity

(2) Not more than 2.8 watts per square centimeter (W/cm^2) (18 watts per square inch) heating capacity

(3) Thermal balance design or thermostat control providing for uninterrupted operation

(4) Provision for mounting below minimum service oil level.

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* (c) An in-line fuel warmer or preheater unit shall be provided on diesel engine driven vehicles to prevent clogging of fuel filters due to wax crystallization in the fuel. The unit shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°C (-40°F) to an outlet temperature of -13°C (+9°F), with a fuel flow rate not less than the maximum fuel demand of the engine fuel system. A coolant shutoff valve shall be provided for the coolant inlet side of the fuel warmer unit. The unit shall not cause heating of the fuel above 27°C (80°F) under any possible condition.

* (d) An in-tank fuel warmer or preheater unit shall be provided on diesel engine driven vehicles. The unit shall use engine coolant to transfer heat to the fuel in one fuel tank. The warmer shall not cause heating of any fuel above 27°C (80°F) under any possible condition, shall not disable or cause elimination of the fuel gage sending unit and shall not violate 3.2.1 or any Federal Motor Carrier Safety Regulation. A coolant shut off valve shall be included. The units required by 3.4.1.9(c) and (d) may be combined.

* 3.4.1.10 Fuel fired engine preheater. When specified (see 6.2), a diesel fuel fired engine water heater shall be furnished to preheat the engine. The heater shall include a timer, a thermostat and a circulating pump and shall be connected directly to the engine coolant system. The heater shall be capable of starting and operating at -51°C (-60°F) and shall heat the engine to +4°C (+40°F) from -51°C (-60°F) in not more than 1 hour. The system shall be equipped with a light, visible to the driver, to indicate that the preheater is operating. The system shall include fuel warmers or preheaters conforming to 3.4.1.9(c) and (d).

* 3.4.1.11 Vernier throttle control. When specified (see 6.2), a manual vernier controlled throttle with quick release shall be furnished, except on electronically controlled engines.

3.4.2 Electrical system. The electrical system shall be in accordance with Federal Motor Carrier Safety Regulations 393.27 through 393.31 and 393.33.

* 3.4.2.1 Starting system. For diesel engine driven vehicles, a 12- or 24-volt direct current (dc) starting system, with a 12-volt dc lighting system shall be furnished. Engine starting equipment shall include an ether starting system, glow plug or grid heater. If an ether system is furnished in lieu of a glow plug or grid heater, it shall be of the measured shot type. The measured shot type ether system shall be key operated or manually operated from the driver's compartment, and shall be inoperative with the engine warm. Complete provisions for a replaceable ether reservoir of not less than 12 fluid ounces shall be furnished. A reservoir need not be furnished. When specified (see 6.2), the starting motor shall be equipped with a thermostat controlled, automatic resetting

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circuit breaker that will protect the motor from overcrank heat damage. The circuit breaker shall not limit cranking ability for a startable engine.

* 3.4.2.2 Ignition system. For gasoline engine driven vehicles, a 12-volt dc ignition system shall be furnished.

* 3.4.2.3 Alternator. Unless otherwise specified (see 6.2), a minimum 85-ampere alternator shall be furnished. The alternator output with the engine at idle speed shall be not less than 45 amperes.

* 3.4.2.4 Lighting. All vehicle lights, reflectors, and wiring shall conform to Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19, 393.20 and 393.22 through 393.26(d). Lights and reflectors shall not be mounted on vertical surface of rub rails or mounted on vehicle bumpers (unless recessed and fully protected). Positioning and guarding shall permit normal replacement of the bulbs and lenses. When right-hand drive is specified by acquisition documents, left dip headlights shall be provided. Left-dip headlights may be provided as a replacement set, stowed in the cab for shipment.

3.4.2.5 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. On class 10 vehicles, front turn signal lights may be integral with parking lights. Turn signal units shall be installed in accordance with SAE J588. Turn signal operating units shall have a visible flash indicator.

3.4.2.6 Batteries. Each battery shall be of 12-volt potential. The total reserve capacity ratings and the total cold cranking ratings at -18°C (0°F), both measured in accordance with SAE J537, shall be not less than specified in table VIII for the indicated engine type. The batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131.

Table VIII Batteries.

Engine type	Reserve capacity (minutes)	Cold cranking (amperes)
Gasoline	100	450
* Diesel (class 10 wrecker only)	180	1,080
Diesel engine, 200 kilowatts (kW) gross (270 gross horsepower) or less	480	1,875
Diesel engine (over 201 kW gross (271 gross horsepower	640	2,500

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3.4.2.7 Auxiliary 24-volt system with trailer receptacle. When specified (see 6.2), an auxiliary 24-volt system with a trailer receptacle shall be furnished. Either a converter type (see 3.4.2.7.1) or an alternator type (see 3.4.2.7.2) system meeting specified requirements shall be furnished. A trailer receptacle, conforming to MS 75021-2, with cover assembly, shall be provided in an accessible location on the rear end of the vehicle. A twelve-conductor truck tractor cable, not less than 2800 mm (110 inches) long, with both ends of the cable equipped with connectors conforming to MS 75020-1 and MS 75020-2, shall be furnished. The cable assembly shall be stowed in the vehicle. The 24-volt, 12-ampere output service lighting circuit shall be connected through the appropriate lighting controls to terminals B, D, E, J and L of MS 75021-2.

* 3.4.2.7.1 Converter type 24-volt system. The 12- to 24-volt converter(s) shall operate from the 12-volt battery (see 3.4.2.7). The output capacity shall be not less than 24 amperes. More than one converter may be provided to furnish a total of 24 amperes.

3.4.2.7.2 Alternator type 24-volt system. The alternator type 24-volt system shall be separate from the 12-volt vehicle lighting and ignition system and shall include:

(a) Nominal 24-volt alternator with not less than 24-ampere rated capacity and capable of providing not less than 7 amperes dc output at normal engine idle speed
(b) Two 12-volt batteries with a combined capacity of at least 40 ampere hours at a 20-hour rate or one 24-volt battery with a 20-ampere hour capacity at a 20-hour rate
(c) Voltage regulating device
(d) An ammeter for the 24 volt system, mounted on the instrument panel.

* 3.4.2.8 Radio interference suppression. The vehicle shall be suppressed to limit the electromagnetic radiation in accordance with SAE J551. Any body equipment emitting electromagnetic radiation shall be suppressed to the same level as the vehicle chassis.

3.4.3 Fue1 system. The fuel system shall conform to Federal Motor Carrier Safety Regulations 393.65 and 393.67.

3.4.3.1 Air cleaner. An air cleaner shall be furnished.

* 3.4.3.2 Fuel tank(s). Fuel tank(s) for class 10 wreckers shall have not less than 75.6 L (20 gallons) total capacity. All other classes of wreckers shall be equipped with step type fuel tank(s) of not less than 189 L (50 gallons) total capacity (163 L (43 gallons) for vehicles destined for California). Except for class 10, when more than one tank is furnished on diesel engine driven vehicles, means shall be provided to assure an equalized fuel level in both tanks. When more than one tank is furnished on gasoline engine driven vehicles, or on class 10 vehicles (diesel or

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gasoline), a selector valve connecting either tank to the engine fuel intake shall be provided and means shall be provided to monitor the fuel level of either tank from a single gage; or an equalizing pump shall be used to maintain the same fuel level in both tanks. Except for class 10 wreckers, fuel tanks shall not extend beyond the rear of the cab.

* 3.4.3.3 Fuel and water separator. The manufacturer's standard or optional fuel filter shall be provided. When specified (see 6.2), a fuel and water separator shall be furnished. The separator shall include a water coalescer and a drain valve. When a separator is required, a combination filter/separator unit may be furnished.

3.4.4 Exhaust system. The exhaust system shall conform to Federal Motor Carrier Safety Regulation 393.83. Vertical exhaust mufflers, if furnished and if capable of being reached easily by personnel entering or leaving either side of the cab, shall be furnished with a heat shield. Vertical exhaust systems, if furnished, shall be provided with a hinged rain cap.

* 3.4.4.1 Spark arresting muffler. When specified (see 6.2), a spark arresting muffler shall be furnished, and shall have an 80 percent arresting efficiency when rated in accordance with SAE J350.

3.4.4.2 Exhaust system safety shield. The exhaust system or hydraulic system shall be shielded so that, in the event of any hydraulic line rupture or any hydraulic fluid leak, the hydraulic fluid will not come into contact with any part of the vehicle exhaust system that is to the rear of the cab.

3.4.5 Gear train.

* 3.4.5.1 Transmission. Unless otherwise specified (see 3.4.5.3), a manually shifted transmission shall be provided. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. Gear ratios in transmission and axles shall be matched to provide a progressive shifting pattern throughout the complete range, and shall provide required vehicle performance (see 3.3 through 3.3.1.3). The transmission shall be provided with power takeoff opening(s) conforming to SAE J704. The transmission shall provide for maximum ease of shifting in all speeds, and the net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine. The transmission shall be the manufacturer's standard design with not less than the number of vehicle forward speeds specified in table IX for the corresponding vehicle class. A multispeed transmission or an auxiliary transmission shall be provided. Auxiliary transmissions shall be of the air shift type.

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TABLE IX. Minimum number of vehicle forward speeds.

Vehicle class	Minimum number of forward speeds	
	4x2 and 6x4 models	4x4 and 6x6 models
10	4	8
21, 24, 28	5	9
32	8	9
34.5, 36, 39.5	9	9
44.5, 50	13	9

* 3.4.5.2 Clutch. The clutch shall be the largest capacity clutch offered for the class of vehicle and the engine furnished, with the clutch torque capacity exceeding the maximum delivered engine torque. The clutch lining shall be asbestos free. The clutch on model 6x4 and 6x6 shall have spring dampening and a greaseable bearing.

* 3.4.5.3 Automatic transmission. When specified (see 6.2), the vehicle shall be provided with an automatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than three forward speeds for class 10, and four forward speeds for all other classes, except model 6x4 and 6x6 shall have five forward speeds. Normal driving range selector position, except for class 10, shall provide not less than four gear ratios without movement of the selector. Except for class 10, the transmission shall be provided with a power takeoff opening.

3.4.5.4 Transfer case. Model 4x4 and 6x6 vehicles shall be provided with a transfer case. Unless the transfer case is equipped with devices which compensate for differential torque and speeds between the front and rear axles, the transfer case shall provide for driver selection of either two-wheel or four-wheel drive for model 4x4 or either four-wheel or six-wheel drive for model 6x6. When furnished, inter-axle compensating devices shall provide for positive transfer of power to all driving axles. Unless otherwise specified, either a one- or two-speed transfer case may be furnished. When specified (see 6.2), a two-speed transfer case shall be furnished. When a two-speed transfer case is furnished, the speedometer shall read accurate speed km/h (mph) with the speed selector in high or low range.

3.4.5.5 Power takeoff. A transmission or transfer case mounted power takeoff shall be provided, except on class 10 vehicles with an automatic transmission. The power takeoff shall be of a rated capacity to operate powered equipment. Controls to operate the power takeoff shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading

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"DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be installed so as to be readily visible to the seated driver (see 3.1.1.13). For class 10 vehicles with an automatic transmission, a split shaft power takeoff or an engine driven hydraulic system with an electric, asbestos free, clutch shall be provided.

3.4.6 Driveline components. Driveline components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

* 3.4.7 Frame. The truck frame reinforcement or heavy duty frame shall be in accordance with the wrecker body manufacturer's recommendation for the size of wrecker unit furnished. Chassis frame rails shall not project beyond the rear end of the body. For all but class 10 wreckers, the chassis shall have a heavy duty main frame or frame with reinforcement extending at least from the rear of the front suspension rear hanger bracket to the bogie trunnion mounting bracket. The heavy duty frame shall have frame rails of greater section modulus than the manufacturer's standard for the class of vehicle furnished and shall provide structural strength at least equivalent to the reinforced frame specified herein. Reinforcement shall provide sufficient structural strength in the chassis frame through increased resisting bending moment, to at least equal the loads imposed with the vehicle loaded to the most severe loading configuration allowed by table X, XII or XIII.

3.4.8 Suspension. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to that specified in 3.2.5. When spring capacity is rated at the spring pads, unsprung weight shall be deducted.

* 3.4.9 Axles. Axle ratings shall be at least equal to that specified in 3.2.5. The front and rear axles on model 4x2 and 6x4 and the rear axles on model 4x4 and 6x6 shall have oil lubricated wheel bearings and axle spindles. Provisions for venting or withstanding pressure buildup and for replenishing the oil supply shall be provided.

3.4.9.1 Rear bogie. The rear bogie on model 6x4 and model 6x6 vehicles shall be of the four-wheel type, complete with axles, springs, torque rods, and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles and a manually or automatically controlled lockout assuring equal power to each rear axle. The manual lockout control shall be located in the truck cab. The axle gear ratios shall provide the performance specified in 3.3 through 3.3.1.3.

3.4.9.2 Two-speed axles. When specified for model 4x2 or 6x4 (see 6.2), a two-speed axle shall be furnished, equipped with electric, vacuum, or air shift and provided with ratios which will permit proper gear splitting. The gear ratios shall provide the performance specified in 3.3 through 3.3.1.3.

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3.4.9.3 Traction control. When specified (see 6.2), the vehicle shall be furnished with traction control on the rear axle (either rear axle for model 6x4 and 6x6). The traction control shall actuate automatically to insure that power is transmitted to the wheel having traction when the opposite wheel loses traction. Maximum traction capabilities shall be maintained at all times under each drive wheel for the life of the vehicle.

* 3.4.10 Wheels, rims, tires and tubes. Unless wide base tires are specified, the vehicle shall be equipped with single front and dual rear wheels. Except as specified herein, rims and tire ratings shall conform to Tire and Rim Association or European Tyre and Rim Technical Organisation recommendations for the type and size of tires furnished. Except when tube type tires are specified in acquisition documents, multi-piece rims shall not be furnished on front axles rated at 6350 kg (14,000 pounds) or less and on rear axles rated at 10430 kg (23,000 pounds) or less. Rims and disc wheel sizes shall be the same for all wheels on each vehicle, except for class 50. Tire size and load range (ply rating) shall be the same for all tires on the vehicle, except for class 50. When specified (see 6.2), wide base type tires and wheels for the front and rear axles shall be provided in lieu of conventional front and dual rear wheels and tires. Wide base wheels shall be interchangeable without the use of an adapter. When specified (see 6.2), disc type wheels shall be furnished.

* 3.4.10.1 Tires. Tires shall be steel belted radial or, when specified (see 6.2), bias ply. Unless otherwise specified, low profile or standard profile tires may be furnished. When specified (see 6.2), low profile tires shall be furnished. Tires shall be tubeless type with highway tread for model 4x2 and 6x4 and with nondirectional mud and snow tread for model 4x4 and 6x6 vehicles. Tires shall be of rated capacity at least equal to that specified in 3.2.5. Tires shall conform to Tire and Rim Association or to the European Tyre and Rim Technical Organisation recommendations.

* 3.4.10.2 Inner tube. When tube type tires are specified in acquisition documents, inner tubes shall be of the heavy duty type, and shall be of the proper size for the tires furnished. Tube type radial tires shall have radial tire inner tubes. Tire flaps shall be provided for tube type tires in accordance with Tire and Rim Association recommendations.

* 3.4.10.3 Carrier for spare tire assembly. A spare tire carrier shall be installed in a readily accessible location on the wrecker. Threaded fasteners, when used to secure a spare tire in the carrier, shall be constructed of or plated with corrosion-resistant material. A vertical or horizontal carrier may be furnished. With the spare removed from the carrier, the deck of the wrecker shall be free of safety obstacles. The carrier design shall enable removal or mounting of the spare wheel assembly using only the tools specified in 3.4.17.1. The carrier shall enable the safe removal and installation of the spare tire assembly from and to the vehicle and carrier without personnel positioning themselves or any part of their body under the spare tire assembly.

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3.4.10.4 Spare wheel or rim. When specified (see 6.2), a spare wheel or rim shall be furnished. When a spare tire assembly is specified (see 3.4.10.5), a spare wheel or rim shall be furnished.

3.4.10.5 Spare tire assembly. When specified (see 6.2), a spare tire assembly shall be furnished. The spare tire assembly shall include an inflated spare tire mounted on the spare wheel or rim. The spare tire shall be of the same size, tread design and load range (ply rating) as the tires furnished on the front axle of the vehicle.

3.4.10.6 Tire chain clearance. Tire chain clearance in accordance with SAE J683 shall be provided. Allowance for spring deflection shall be included.

3.4.11 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43 and 393.45 through 393.52. The vehicle shall be equipped with full air brakes, except that class 10 vehicles shall be furnished with power-assisted hydraulic brakes. Brakes shall be furnished on all wheels. Brake linings shall be of nonasbestos material on all front axles, on single rear axles rated at 10 430 kg (23,000 pounds) or less and on tandem rear axles rated at 20 900 kg (46,000 pounds) or less.

3.4.11.1 Split hydraulic brake system. The power-hydraulic brake system shall be so arranged as to provide separate systems for at least two wheels and so designed and constructed that rupture or leakage type failure of any single pressure component of the service brake system, except structural failures of the brake master cylinder body, effectiveness indicator body, or other housing common to the divided system, will not result in complete loss of function of the vehicle brakes when force on the brake pedal is continued. “Pressure component” means any internal component of the brake master cylinder or master control unit, wheel brake cylinder, brake line, brake hose, or equivalent, except power assist components.

3.4.11.1.1 Indicator light. The split hydraulic brake system shall be equipped with an electrically operated red light mounted on the instrument panel to indicate system effectiveness. The light shall have an area of not less than 125 square millimeters (mm^2) (0.196 square inch). It shall illuminate before or upon application of the brakes when an actuating-pressure component of the system has sustained a loss of pressure. The indicator light system shall include a means for testing by the vehicle operator to assure that the light bulb is operable.

3.4.11.2 Air brakes. Air brakes shall be complete with all necessary components, including the following:

(a) Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 205 L/min (7-1/4 cubic feet per minute (cfm)) for model 4x2 and 4x4 and not less than 340 L/min (12 cfm) for model 6x4 and 6x6

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(b) Air storage reservoir(s); each tank shall be equipped with a drain, and with safety and check valves between the compressor and the last reservoir tank
(c) Foot control, suspended or treadle type
(d) Air control valves
(e) Air pressure gage, visible to the driver
(f) Low air-pressure warning, visible and audible
(g) Service brake stop lamp switch
(h) Automatic moisture ejector
(i) Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes (except on models 4x4 and 6x6).

3.4.11.2.1 Air dryer. A replaceable cartridge desiccant type air dryer shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve.

3.4.11.3 Trailer brake control system. In addition to the components specified in 3.4.11.2 and 3.4.11.2.1, a trailer brake control system shall be furnished, except on class 10 and class 21 vehicles. The trailer brake control system shall include:

(a) Identification of emergency and service lines
(b) Coincident control of trailer brakes with prime mover foot control
(c) Independent hand control for trailer brakes
(d) Prime mover protection valve with dash control and automatic break-away feature
(e) Trailer stoplight operable with the foot brake and with a hand control for trailer brakes
(f) Two SAE J844 coiled air hoses, not less than 2800 mm (110 inches) long, with SAE J318 gladhand couplers on both ends of hoses. The hoses shall be packaged and stowed in the vehicle tool compartment for shipment
(g) Air connectors for a trailer with SAE J318 gladhand couplers mounted at the rear of the vehicle, located to prevent interference with a trailer
(h) Dummy gladhand couplers with chains
(i) Prime mover only parking brake valve to permit prime mover parking brakes to be applied while charging the trailer air brake system (with air or air-hydraulic brakes).

3.4.11.4 Brake controls for use from a towing vehicle. When specified (see 6.2), except for class 10, the vehicle shall be furnished with a system for controlling the brakes from a towing vehicle (a second wrecker). The installation shall be complete with air brake couplers, relay emergency valve with no-bleed-back feature (except when spring applied emergency brakes are

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furnished), additional air lines and fittings. The service and emergency couplers shall be mounted on the front in a protected position providing for ready attachment of air hoses from a towing vehicle. The service and emergency couplers shall be identified and provided with dummy gladhand couplers with chains. The system shall not compromise conformance to any Federal Motor Carrier Safety Regulation referenced herein or to any Federal Motor Vehicle Safety Standard.

* 3.4.11.5 Increased braking capability. When specified for class 32 and larger diesel engine driven vehicles (see 6.2), the service brakes shall be augmented by one of the following braking systems:

(a) An electromagnetic driveshaft retarder
(b) A system which opens all or some of the engine exhaust valves near the end of the compression stroke, thereby covering vehicle motion to a pumping loss
(c) A controlled gate valve in the exhaust manifold, which produces back pressure on the engine pistons during the exhaust stroke
(d) When an automatic or semiautomatic transmission is specified, a hydrodynamic retarder integral with the transmission.

A dash mounted switch shall be provided to activate, modulate, or cut out the brake augmentation. The switch shall be marked to indicate its position. When active, the system shall be fully controlled by means of the conventional driving controls to apply retardation during vehicle deceleration, and to cut it out in the other operating modes. Each of the four systems shall permit the wrecker loaded to the required GVW to negotiate a 6 percent downgrade without exceeding 44 km/h (27 mph), within the operating restrictions of the engine and transmission and without the use of service brakes. For (b) or (c), above, the retarder shall be approved by the engine manufacturer.

* 3.4.12 Cab. Unless otherwise specified, the chassis manufacturer shall furnish any type of his standard or optional full width cab. When specified (see 6.2), except for class 10, a cab with a forward tilting hood and fender assembly, including tilting and locking mechanism shall be furnished. Tilting shall not interfere with present or future installations of additional equipment, such as a snowplow or a front mounted winch. When specified (see 6.2), except for class 10, a tilt cab with tilting and locking mechanism shall be furnished. Tilt cab features shall be in accordance with the Employee Safety and Health Standards of Federal Motor Carrier Safety Regulation 399, including the first step height, which shall be not more than 610 mm (24 inches). Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, with both doors equipped with external, key-operated locks. Drip rails shall be installed above the cab doors. When the front tires extend beyond the cab fenders, rubber fender extensions extending at least to the outside of the tire tread shall be furnished. Tilt cabs, with a height from the ground to the top of the windshield of more than 2030 mm (80 inches), shall include

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provisions to facilitate cleaning the windshield. Provisions shall consist of a bumper step or bumper step cutouts and a grab handle located under the windshield.

* 3.4.12.1 Cab interior. The cab shall have an upholstered, full width adjustable seat and back or an individual, adjustable driver's seat and an individual passenger seat. The color of the upholstery and the interior finish shall be compatible with the exterior color (see 3.1.1.1). White upholstery shall not be furnished. Interior lighting shall be provided. Two pairs of seat belts shall be installed. Outboard seats shall have a combination pelvic and upper torso restraint seat belts.

3.4.13 Steering. Power steering shall be furnished.

* 3.4.14 Windshield wipers and washers. The vehicle shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be of the multispeed type and operated by either air or electric motor(s). When specified (see 6.2), and if electric motor wipers are furnished, intermittent wipers shall be furnished.

3.4.15 Front bumper. The front bumper shall be as specified in 3.5.3 or 3.6.3, or when a front winch is specified, 3.4.26.2.

3.4.16 Rear end protection. Rear end protection shall be as specified in 3.5.4 or 3.6.4.

* 3.4.17 Too1 stowage. Stowage space of sufficient size to accommodate a vehicle jack, hand tools, anti-skid chains (for outside tires on duals only) and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. The stowage space shall provide for positive retainment of this equipment during vehicle operation. Stowage space for these tools may be furnished in the cab. When stowage space for these tools is located outside the cab, it shall be weatherproof and shall provide for locking.

3.4.17.1 Tools. When specified (see 6.2), the vehicle shall be furnished with tools required for exchanging a mounted tire assembly with the spare assembly, and shall include at least a jack, a jack handle and a wheelnut wrench. The jack shall be of such closed height as to permit its location under the axle, or other satisfactory lift point, at any wheel with the tire flat. The jack, without blocking, shall be capable of raising any wheel of the loaded vehicle to a height adequate to permit removal and replacement of a wheel and tire assembly.

3.4.18 Heater and defroster. A hot water heater shall be provided. The heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to the floor and to defroster louvers. The heater shall be complete with blower and mounted controls convenient to the driver.

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3.4.19 Controls and operating mechanisms. All controls and operating mechanisms shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of the operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted. All instruments shall be visible to the driver when seated in the driving position.

3.4.20 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

(a) Key operated ignition switch
(b) Ammeter, charging indicator or voltmeter
(c) Fuel gage
(d) Oil pressure gage or red indicator warning light
(e) Engine temperature gage or high coolant temperature or low coolant level red indicator warning light
(f) Except for class 10, a high coolant temperature or low coolant level alarm buzzer
(g) Speedometer with recording odometer
(h) Ash receptacle
(i) Dual sunvisors
(j) Driver's compartment ventilator other than window
(k) Tachometer.

* 3.4.21 Rearview mirrors. Outside rearview mirrors shall be mounted on both sides of the cab. The mirrors shall have flat and convex areas. The flat portion shall have not less than 320 cm^2 (50 square inches) of reflective area. The convex portion shall have not less than 155 cm^2 (24 square inches) of reflective area and a radius of curvature of not less than 510 mm (20 inches). The convex portion shall be attached to the lower mirror supporting arm and shall not interfere with use of the flat mirror. The mirrors shall have not less than two supporting arms. When specified (see 6.2), the curbside flat mirror shall be of the motorized type, with remote control. The mirror motor shall provide not less than 60 degrees horizontal rotational viewing range. When specified (see 6.2), the flat mirrors shall be electrically heated. Mirror remote and heating controls shall be within reach of the seated driver.

3.4.22 Horn. The manufacturer's standard electric horn shall be furnished. When specified (see 6.2), except for class 10, in addition, an air operated horn shall be furnished.

3.4.23 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous

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trouble-free performance under severe operating conditions. The engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.4.24 Back-up alarm. The vehicle shall be provided with an audible, pulsating, signaling device (electric or mechanical) to caution personnel when the vehicle is in reverse gear operation.

3.4.25 Air conditioning. When specified (see 6.2), the vehicle shall be equipped with the manufacturer's standard air conditioning system. The air conditioning system shall include tinted windshield and tinted glass.

3.4.26 Front mounted winch. When specified (see 6.2), except for class 10, a winch assembly, mounted on the front of the vehicle and powered by the manufacturer's standard power takeoff or powered hydraulically, shall be provided. The winch shall conform to SAE J706. The winch shall be of the single drum type and shall have one forward and one reverse speed. All winch controls shall be mounted inside the cab. Winch controls shall be located to provide no interference with the entrance or exit of the driver. The winch shall have a rated single line pull capacity of not less than 89 kN (20,000 pounds) pull on the bare drum. The winch line speed on the bare drum shall be between 4.6 and 9.1 meters per minute (m/min) (15 and 30 feet per minute) at an engine speed equal to 50 percent of engine governed speed. The winch shall be wound with not less than 56 m (185 feet) of 16 mm (5/8-inch) diameter, preformed, 6x37, improved plow steel, independent wire rope core (IWRC) and shall be equipped with an end chain and hook. An integral, adjustable, automatic safety brake shall be provided. The winch shall be equipped with a roller guide. Angles of the winch drive line U-joints shall be not greater than 16.5 degrees.

3.4.26.1 Winch drum guard. A winch drum guard shall be furnished. The guard shall confine the cable to the area between the drum flanges. The guard shall consist of not less than 6.4 mm (0.25 inch) vertical side plates, conforming to the outside radius of the drum flanges. Six bars, 9.5 mm (0.375 inch) by 32 mm (1.25 inches), shall be welded to the vertical side plates. Three bars shall be located on the top and spaced equally on the top radius, and three bars shall be located on the bottom and spaced equally on the bottom radius. The vertical distance between the vertical side plates and the drum flanges shall be not more than half the specified cable diameter.

* 3.4.26.2 Front bumper. When a front mounted winch is specified (see 3.4.26), the bumper shall be mounted forward of the winch. The open area on either side of the winch shall be covered with a combination step plate and gravel guard. The bumper shall be of either the channel type or pipe type. When a pipe type front bumper is furnished, the nominal diameter

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shall be not less than 75 mm (three inches), and the wall thickness shall be not less than that specified in schedule 40 of ASTM A53. The pipe type front bumper shall have half-round ball ends.

* 3.4.26.3 Combination step plate and gravel guard. When a front mounted winch is specified, a combination step plate and gravel guard shall be furnished. The step plate shall be fabricated of not less than 14 gage (1.897 mm) (0.0747 inch) steel tread plate exclusive of projections. The step plate shall be secured to the front bumper and shall be not less than 6.4 mm (0.25 inch) or not greater than 9.5 mm (0.375 inch) from the chassis sheet metal. Rubber flaps may be employed to achieve these dimensions and shall be employed when required by tilting hood/fender vehicles for unrestricted tilting. The step plate shall be capable of supporting 1460 kilograms per square meter (kg/m^2) (300 pounds per square foot). The step plate shall not deflect more than 3.2 mm (0.125 inch) under the loads imposed.

* 3.4.27 AM/FM radio. When specified (see 6.2), the manufacturer's standard AM/FM radio shall be provided.

* 3.4.28 Servicing and adjusting. Prior to acceptance of the vehicle by the Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use including at least the following: alinement of lights; adjustment of engine and brake systems; filling and charging of batteries; alinement of front wheels; inflation of all tires; complete lubrication of chassis, engine, running gear, and mounted equipment with grades of lubricants recommended for the ambient air temperature at the delivery point; servicing of cooling system in accordance with 3.4.28.1; servicing of windshield washer reservoir with water and appropriate additives; and checking level of hydraulic fluid reservoir.

* 3.4.28.1 Engine coolant. The engine coolant shall be a solution of ethylene glycol antifreeze and water or propylene glycol antifreeze and water, in equal parts of antifreeze and water by volume. Ethylene glycol antifreeze shall conform to ASTM D 4985 with not more than 250 parts per million silicates. Propylene glycol antifreeze shall be in the specific formulation approved by the engine and cooling system manufacturers. When specified (see 6.2), the percentage of antifreeze in the cooling system shall be increased to provide protection against freezing down to -54°C (-65°F).

* 3.5 Type I hydraulic wrecker. Type I hydraulic wrecker shall have the dimensions and load capacities specified in table X for the corresponding vehicle size and GVW.

* 3.5.1 Type I hydraulic wrecker optional extended boom. When specified for size 40T and 45T (see 6.2), extended boom length ("j" in table X) shall be not less than 3960 mm (156 inches) with an extended boom rating ("h" in table X) of not less than 5450 kg (12,000 pounds).

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TABLE X. Type I hydraulic wrecker dimensions and load capacities

Characteristic	Vehicle						
Vehicle size	5T	7T	20T	25T	35T	40T	45T
GVW kg (pounds)	4540 (10,000)	4540 (10,000)	12 700 (28,000)	14 500 (32,000)	17 900 (39,500)	17 900 (39,500)	22 700 (50,000)
a. Basic rating, lift capacity (tons) not less than	5	7	20	25	35	40	45
b. Lift-tow capacity kg (pounds) not less than	815 (1,800)	815 (1,800)	3400 (7,500)	3630 (8,000)	3630 (8,000)	3630 (8,000)	3630 (8,000)
c. Centerline of rear axle to rear of tow bar mm (inches)	1700±75 (67±3)	1700±75 (67±3)	1700±75 (67±3)	1700±75 (67±3)	--- ---	--- ---	--- ---
d. Centerline of bogie to rear of tow bar mm (inches)	--- ---	--- ---	--- ---	--- ---	2490±75 (98±3)	2490±75 (98±3)	2490±75 (98±3)
e. Wheelbase mm (inches) not less than/not more than	3020 (119)	3020 (119)	4600 (181)	4600 (181)	4700/5310 (185/209)	4700/5310 (185/209)	4700/5310 (185/209)
f. Boom rating:							
g. Fully retracted (pounds) not less than	4540 (10,000)	6350 (14,000)	18 150 (40,000)	22 700 (50,000)	31 750 (70,000)	36 300 (80,000)	40 800 (90,000)
h. Extended to not less than (j) below not less than kg (pounds)	2270 (5,000)	2270 (5,000)	5450 (12,000)	9070 (20,000)	10 900 (24,000)	13 600 (30,000)	13 600 (30,000)
i. Boom geometry:							
j. Extended mm (inches) past rear plate not less than	1620 (64)	1620 (64)	1830 (72)	2440 (96)	2440 (96)	2540 (100)	2540 (100)
k. Elevation (degrees) not less than	45	45	45	45	45	45	45
l. Winch rating:							
m. Each winch kg (pounds) not less than	3630 (8,000)	3630 (8,000)	9070 (20,000)	11 350 (25,000)	15 850 (35,000)	18 150 (40,000)	20 400 (45,000)
n. Winch cable class	6x19	6x19	6x19	6x19 or 6x25	6x19 or 6x25	6x19 or 6x25	6x19 or 6x25
o. Cable size mm (inches) not less than	10 (3/8)	10 (3/8)	14 (9/16)	16 (5/8)	19 (3/4)	19 (3/4)	19 (3/4)
p. Cable length m (feet) not less than	30.5 (100)	30.5 (100)	61 (200)	53.4 (175)	61 (200)	76.3 (250)	76.3 (250)
q. Cable type	Preformed, extra improved plow steel cable with steel core.						

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3.5.1.1 Type I hydraulic wrecker unit. The wrecker unit shall have a welded steel frame mounting a two- or three-stage power telescoping and power elevating boom and power operated winch(s) as a unit assembly.

3.5.1.2 Lifting capability. The wrecker shall be capable of lifting a load equal to its lifting capacity as specified in 1.2.2. The wrecker shall be capable of extending, retracting, raising and lowering with loads as specified herein. Controls for operating the boom, winches and wheel-lift shall operate smoothly, with the loads specified, to avoid producing shock loads. The boom, winch and wheel-lift assembly shall maintain loads imposed as specified without slippage when controls are released. Snatch blocks shall be used, as required, to prevent exceeding the cable working load limit.

* 3.5.1.3 Type I hydraulic wrecker boom extension cylinders. With the boom fully retracted and positioned within 9 degrees of horizontal, a load equal to 75 percent of the extended boom rating specified in line "h" of table X shall be capable of being extended to the extended boom dimension specified in line "j" of table X.

3.5.1.4 Type I hydraulic wrecker boom elevation cylinder(s) (retracted). With the boom fully retracted and positioned at an angle of 20 degrees above horizontal, the boom elevation cylinder(s) shall be capable of lifting a load equal to 55 percent of the retracted boom rating specified in line "g" of table X.

3.5.1.5 Type I hydraulic wrecker boom elevation cylinder(s) (extended). With the boom in the extended position (see line "j" of table X) and positioned at an angle of 20 degrees above horizontal, the boom elevation cylinder(s) shall be capable of lifting a load equal to the extended boom rating specified in line "h" of table X.

3.5.2 Type I body structure. The body shall be of the deck type with flanged side plates of carbon steel, or of aluminum of equivalent strength. Wheelhousings or fenders shall cover the rear wheels. When the width over the tires exceeds the body width, rubber fenders extending out to the width over the tires shall be provided. The floor shall be steel or aluminum, safety tread diamond plate, reinforced to support not less than 1460 kg/m^2 (300 pounds per square foot) of area. The minimum thickness of the body shall be as shown in table XI for the specified vehicle size. Mounting aids shall be provided on each side of the body or, on each side of the rear, for access to the deck. Means for draining rain water from the deck shall be provided. The deck may have a kick-up over the rear axle and may be slanted toward the ground behind the rear axle; sizes 5T and 7T deck may be slanted from behind the cab to the end of the deck. Hook loops shall be provided at the rear and, except for sizes 5T and 7T, at the sides, all for use when the load is double-reeved with snatch block(s). Lights mounted on the side plates shall be recessed or shall be of the armored type.

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TABLE XI. Body thickness.

Vehicle size	Body thickness
5T, 7T and 16T	14 gage (1.897 mm) (0.0747 inch)
20T through 45T	12 gage (2.657 mm) (0.l046 inch)

3.5.3 Front push bumper. Unless a front mounted winch is specified, a front push bumper shall be provided. The bumper shall be constructed of not less than 4.76 mm (0.187 inch) steel plate for sizes 5T and 7T, and not less than 6.35 mm (0.25 inch) steel plate for all other sizes, all flanged to a depth of 75 mm (three inches) top and bottom, or flanged to a depth of 50 mm (two inches) top, bottom and ends; all shall be not less than 368 mm (14.50 inches) between top and bottom flanges; all shall extend across the full width of the vehicle cab, plus or minus 75 mm (three inches); all shall have rubberized push pads or padding; and all shall be reinforced and mounted to the chassis frame to withstand a push of not less than 22.25 kN (5,000 pounds) on any portion without permanent distortion. When the front bumper is mounted on the front spring hanger brackets, the brackets shall be reinforced to reduce the shear forces on the bracket-to-frame mounting bolts or rivets.

3.5.4 Backplate. A backplate shall be provided. The backplate shall be constructed of not less than 4.76 mm (0.187 inch) steel plate for sizes 5T and 7T. The backplate shall be not less than 6.4 mm (0.25 inch) steel plate for all other sizes, flanged at the bottom, reinforced and mounted to transfer the loads imposed on the backplate and towing devices directly to the chassis frame. The backplate shall be the full width of the vehicle. The backplate on sizes 20T through 45T shall be not less than 530 mm (21 inches) from the bottom flange to the top edge of the plate. The backplate shall curve forward or shall be offset forward at each end to provide the maximum cornering clearance when towing a vehicle. Taillights, stoplights, and directional signal lights shall be inset in the backplate or shall be in a protected location. Slots for scotch block chains shall be furnished.

3.5.5 Load line winches. Not less than one winch on size 5T and two independent winches on sizes 7T through 45T shall be provided. The winch(s) shall be of the variable speed, hydraulic type. Each winch shall have a drum pull rated capacity with a single line on the bare drum of not less than that specified in line “m” of table X, with a safety factor of not less than 2.0. Each winch shall be provided with a pressure relief valve set at 110 percent of the maximum winch rating. Winches shall conform to or exceed the requirements of SAE J706. Means shall be provided for free unreeling of the load line. Means shall be provided to prevent overtravel of the drum when free unreeling. Load lines shall be equipped with hooks of sufficient capacity for rated loads.

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3.5.6 Controls. At least one throttle control, located on the left side at the rear, shall be provided. An independent control for each hydraulic function of the wrecker shall be provided on both the left and right sides of the wrecker as follows:

(a) An independent control for each winch (forward, reverse, free unreeling and hold) on both sides
(b) Boom elevation, lowering and hold control on both sides
(c) Boom extension, retraction and hold control on both sides
(d) When hydraulic outriggers are furnished, an independent lower, raise and hold control for only the left outrigger on the left side and an independent control for only the right outrigger on the right side
(e) When rear spades are furnished, a control to simultaneously lower, raise and hold both spades; one control on the left side and one control on the right side
(f) Except for size 5T and 7T, when a wheel-lift is furnished, controls for all wheel-lift functions shall be on a portable remote control box with not less than 7.02 m (23 feet) of control cable.

Throttle control head shall be corrosion-resistant twist lock, push button lock, or micro-adjustable mechanism. Push-pull cable controls shall be recessed. When control cables are used, they shall be stainless steel with a neoprene or equivalent outer cover. Control cables or rods shall be routed and anchored to prevent damage and interference with operations by snow and ice. When space is not available in the sides of the vehicle to permit recessing the throttle and winch controls, upright type controls may be furnished at the right and left rear corners of the vehicle. All controls shall be of the deadman type, and shall automatically return to a neutral hold position when released.

3.5.7 Instruction and capacity plates. Operating positions of controls shall be identified on instruction plates, or instructions shall be etched into the control knobs. Plates inscribed with the payload capacity in the lift-tow load position shall be provided, one at both control stations. A plate showing capacity loads for fully retracted and fully extended boom lengths, and for a fully raised and fully lowered boom, shall be attached to the sides of the wrecker at both control stations. All plates shall be corrosion-resistant metal or weather-resistant plastic.

* 3.5.8 Boom assembly. The boom assembly shall be a hydraulically operated telescoping type, capable of single winch operation for size 5T and of single and twin winch operations for sizes 7T through 45T, all with a minimum 120-degree directional boom tip sheave or roller system. Under dynamic conditions the boom shall meet performance requirements without permanent deformation of any component in or along the boom or its supporting structure. The boom shall be capable of elevation to not less than 45 degrees above horizontal. The boom shall be capable of lowering to within 9 degrees of horizontal.

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3.5.81 Frame structure. A welded steel frame shall be provided for the telescoping boom. The frame shall be internally or externally gusseted to withstand rated capacity loads imposed by the boom over the rear of the truck and to withstand side loads from the end of the boom.

3.5.8.2 Boom construction. The boom shall be designed, engineered and constructed of materials that are conducive to the configuration and maximum performance requirements specified herein. Hydraulic lines and other associated parts which are fastened to or inside the boom shall be positioned for inspection, maintenance and replacement and shall have maximum protection against damage.

3.5.9 Forward mounted outriggers. Except on size 5T and 7T, telescopic hydraulic outriggers shall be attached to the frame structure of the wrecker or shall be incorporated in the construction of the wrecker, one at each side, at the forward end of the body. Outriggers shall be adjustable for varying ground distance. The outriggers shall be positioned such that the legs make an angle with the ground surface of approximately 35 degrees. The outriggers shall be positioned on the wrecker in close proximity to the point of pull of a side load (near the side hooks). Each leg shall have a ground pad of not less than 387 cm^2 (60 square inches) in surface area.

3.5.10 Rear spades. Unless otherwise specified (see 6.2), hydraulically operated rear spades shall be furnished, except on size 5T and 7T. The spades shall be provided with removable or folding feet for use on paved surfaces. The spades shall be installed to resist rearward movement of the wrecker when pulling with a cable from the rear.

3.5.11 Lift-tow load. Without the use of counterweights, the wrecker shall have sufficient front end curb weight to provide the lift-tow load specified in line “b,” table X, with not less than one half of the front axle curb weight maintained on the front axle. The actual lift-tow load capacity shall be determined as follows: one half the front axle curb weight of the wrecker multiplied by the wheelbase of the wrecker and divided by the distance to the lift-tow position as defined in table X. The wrecker shall be capable of lifting the front of the wrecker off the ground by use of a load in the lift-tow position, without any damage to the truck wrecker.

3.5.12 Wheel-lift. For sizes 5T, 7T and 20T, and when specified (see 6.2) for size 25T, a wheel-lift/towing device for towing automobiles, light trucks and vans shall be furnished. Size 20T wheel-lift shall also be capable of towing medium duty trucks. Size 25T wheel-lift also shall be capable of towing heavy duty trucks and buses. The wheel-lift shall be capable of lifting and securing a towed vehicle by its wheels for damage-free transport. The wheel-lift shall have a frame, a lifting arm, a cross-bar and two wheel restraints. The lifting arm shall be hydraulically powered for raising, lowering, extending and retracting under load. The lifting arm shall be hydraulically powered to raise, lower and tilt up and down underload. The cross-bar shall be

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manually extendable width-wise to accommodate a wide variety of vehicles. The wheel restraint shall be manually extendable to accommodate various tire sizes. Safety tiedown straps or chains shall be provided for each wheel restraint to secure the towed vehicle to the wheel-lift. The wheel-lift shall be furnished with a welded steel heavy duty frame designed exclusively for the wheel-lift. The wheel-lift shall be self-stowing beneath the body or recessed into the backplate, without interfering with use of the tow hitch. The tow bar shall pivot sufficiently to allow the wrecker to make a 90-degree turn with a vehicle in tow. The wheel-lift shall conform to table XII for the corresponding vehicle size. Lifting and towing capacity shall be with the wheel-lift extended to the position specified in table XII. Extended position shall be measured from the backplate to the centerline of the wheels of a subcompact automobile in place on the wheel restraints. Retracted position shall be measured from the backplate to the rearmost part of the wheel-lift. Size 20T and 25T wheel-lifts shall include frame lift forks and adapters.

TABLE XII. Wheel-lift units.

Vehicle Size	Lifting Capacity minimum, kg (pounds)	Towing Capacity minimum, kg (pounds)	Extended Position minimum mm (inches)	Retracted Position maximum, mm (inches)
5T, 7T	1360 (3,000)	3400 (7,500)	1370 (54)	810 (32)
20T, 25T	4540 (10,000)	11 340 (25,000)	1980 (78)	810 (32)

3.5.13 Accessories. The following accessories shall be provided:

(a) Two snatch blocks with hooks for double reeving. For single winch vehicles, only one snatch block is required
(b) Universal steering wheel clamp
(c) Except for size 5T, a telescoping tow hitch, with rated capacity not less than wrecker GVW. Two tow chains of high-test, coil grade, bright zinc steel with a J-hook on one end and a T-hook and a grab hook on the other end; 11 mm by 2130 mm (7/16 inch by 7 feet) in length for size 7T wreckers, 16 mm by 2440 mm (5/8 inch by 8 feet) in length, for all other sizes; all conforming to SAE Jl142. Crossbeams and spacer blocks in accordance with SAE Jl142 shall also be provided
(d) Two scotch blocks with chains
(e) Two 8 mm (5/16-inch) high-test steel safety chains for size 5T and 7T wreckers, and two 16 mm (5/8-inch) high-test steel safety chains for all other sizes, each with a bright zinc finish, each 3.66 m (12 feet) long, with a clevis at one end for attaching the chain to an eye on the boom and a grab hook with spring-loaded safety latch on the other end for attaching the chain to a vehicle being towed

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(f) Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i)
(g) A steel stowage box equipped with weatherproof hinged cover and provisions for locking, secured in the body; or a built-in side stowage compartment with side or bottom hinged door(s) and a key operated lock. The stowage box cover shall be provided with a retaining device to hold the cover in the fully open position
(h) For sizes 5T, 7T, 20T and 25T, and when specified for other sizes (see 6.2), a towing sling suitable for towing automobiles. The sling shall include reinforced rubber straps to prevent automobile bumper and grille damage. The sling lifting capacity shall be not less than 1585 kg (3,500 pounds). The sling shall function in conjunction with the chains and grab hooks in (c) above.

* 3.5.14 Electrical accessories. The following electrical accessories shall be furnished:

(a) Two swivel type sealed-beam floodlights, with base mounted weatherproof switches installed at the rear of the cab or with a switch located in the cab. An indicator light shall be located on the instrument panel and marked to indicate the function of the indicator light and the cab switch
(b) Undercarriage mounted work light(s), stationary or folding type, providing illumination at the rear of the vehicle. The lens shall be protected to prevent damage when the vehicle is in transit. The light switch shall be located in the cab and marked to indicate the function and position of the switch
(c) An elevated lighting bar containing rotating amber beacon(s) and brake stop lights attached with reinforcement at a point on the vehicle centerline for 360-degree visibility. The rotating amber beacon(s) shall be operated from the vehicle cab by means of a separate switch with an indicator light located on the instrument panel, and shall be marked to indicate the function of the indicator light and the switch
(d) An auxiliary safety light kit for use when towing a disabled vehicle that has no functioning lights. The kit shall be comprised of two stop-tail-turn signals on a square magnetic base, or on a strap-on light bar. 9.14 m (thirty feet) of cable with four conductor plugs and a mating receptacle on the wrecker shall be provided. A 7.6 m (25-foot) extension cable shall also be furnished. The light kit shall be stowed in a plastic storage box
(e) Wiring conforming to Federal Motor Carrier Safety Regulations 393.27 and 393.28.

3.5.15 Hydraulic system. The hydraulic system shall consist of single or tandem type pumps for sizes 5T, 7T and 20T; and tandem pumps for sizes 25T through 50T; hydraulic cylinders, strainer, filter, reservoir, pressure relief valve and all necessary lines, lockouts, restrictors, and

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control valves to insure positive and safe control of all operations and to provide protection in the event of loss of hydraulic power. All components of the hydraulic system shall be positioned or shielded to prevent hazardous fluid contact with heated surfaces or electrical apparatus.

3.5.15.1 Hydraulic pump(s). Hydraulic pump(s) with required pressure and flow capacities to operate hydraulically powered components shall be provided. The pump(s) shall have sufficient capacity to operate all hydraulic components and shall provide for operation of at least three functions simultaneously. The hydraulic pump(s) shall be driven as specified in 3.4.5.5. The pumping system shall include an adjustable pressure relief valve or unloading valve with a range of adjustment which allows raising, lowering, extending and retracting the boom with any working or test load specified herein.

3.5.15.2 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided, with sufficient capacity to hold the required volume of hydraulic fluid, plus not less than 10 percent expansion space. The reservoir shall be provided with a filler opening and a vented filler cap located to facilitate filling the tank from a 3.78 L (one-gallon) container using a funnel. The vented filler cap shall have means to prevent fluid from splashing out of the tank during movement of the vehicle. The filler cap shall be attached to the tank by a corrosion-resistant chain. Provision shall be made for draining the reservoir. The reservoir drain shall be located so that the fluid will not drain on other components of the vehicle, and shall allow recovery of the fluid in a container. The reservoir shall have a corrosion-resistant metal plate indicating the type of hydraulic fluid furnished. The reservoir shall indicate the maximum fluid level in the reservoir with the boom retracted for travel and the hydraulic lines and components full of fluid. A sight level or dip stick gage shall be furnished to enable the level of the fluid in the reservoir to be observed. The tank shall be clean.

3.5.15.3 Hydraulic fluid. Unless otherwise specified, hydraulic fluid conforming to the manufacturer's standard or recommended requirements shall be provided in the hydraulic system. When specified (see 6.2), hydraulic fluid conforming to MIL-H-83282 shall be provided in the hydraulic system. A corrosion-resistant metal plate located near the hydraulic system fill opening shall be furnished stating "CAUTION: USE HYDRAULIC FLUID CONFORMING TO MIL-H-83282 ONLY."

3.5.15.4 Oil filter and strainers. A full-flow type, 10-micron discharge oil filter in the return line to the reservoir shall be provided. A 100-mesh suction line strainer in the suction line from the reservoir shall be provided. The filter and the suction strainer shall be easily accessible for periodic cleaning, or for replacement of the filter cartridge. The filter and the strainer shall have sufficient capacity to allow the pump to deliver the rated flow required for all operations specified herein.

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3.5.15.5 Hydraulic fluid temperature. The hydraulic fluid temperature shall be maintained within safe limits to prevent seal damage or approach to the flash point of the hydraulic fluid during all operating conditions of the vehicle and use of the boom and winches for continuous periods up to one hour. Excessive heating (more than 13°C (55°F) above ambient) of the fluid shall be unacceptable under any operating condition. Provisions shall be made to prevent malfunction or unactuated operation of the hydraulic system due to changes in fluid temperature.

3.5.15.6 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals with rod wipers to prevent the entrance of moisture and dirt into the cylinders, and leaks of hydraulic fluid from the cylinders. Preformed packings shall be used in the cylinders where static seals are required. The cylinders shall be compatible with the hydraulic system and shall have sufficient cross-sectional area to raise and hold the boom at the rated capacity without the use of pins or other supporting devices. Valves shall be included in the hydraulic system to prevent load drift. All hydraulic cylinders for raising and lowering the boom and outriggers, and for extending and retracting the boom for sizes 25T through 45T, shall be provided with pilot check valves, holding valves or counterbalance valves to prevent descent of the boom assembly, retraction of the boom, and retraction of the outriggers in the event of a loss of hydraulic pressure. Means shall be provided to install pressure gages for testing hydraulic system pressure and relief settings. All surfaces of hydraulic ram piston rods shall be chrome plated to a thickness sufficient to withstand a 30-hour salt spray test in accordance with ASTM B117.

3.5.15.7 Hydraulic lines and fittings. Hydraulic lines shall be located for maximum ease of removal and installation, visual inspection for failure, and maximum protection from damage during operation of the boom assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered and where installation of components requires a flexible coupling. Hydraulic lines and fittings shall be air- and liquid-tight and shall be installed to prevent damage from vibration and chafing. The system shall be designed to prevent hydraulic transients. Provisions shall be made for bleeding air from the hydraulic system and for completely draining all hydraulic lines when necessary. Hose assemblies and all seamless steel hydraulic lines shall be adequate to provide the necessary rate of flow, temperature and pressure required to operate the system smoothly and efficiently. Galvanized pipe and copper tubing shall not be acceptable.

3.6 Type II mechanical wrecker. Type II mechanical wreckers shall have the dimensions and load capacities specified in table XIII for the corresponding vehicle size and GVW.

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TABLE XIII. Type II mechanical wrecker dimensions and load capacities.

Characteristic	Vehicle model and GVW							
Size	16T	24T	24T	24T	24T	24T	24T	24T
Model	4x2	4x2	4x2	6x4	6x4	4x4	6x6	6x6
*GVW kg (pounds)	9530 (21,000)	10 900 (24,000)	14 500 (32,000)	15 650 (34,500)	20 200 (44,500)	10 900 (24,000)	16 350 (36,000)	20 200 (44,500)
a. Basic rating, lift capacity kg (tons), not less than	14 500 (16)	21 750 (24)	21 750 (24)	21 750 (24)	21 750 (24)	21 750 (24)	21 750 (24)	21 750 (24)
b. Lift-tow capacity kg (pounds), not less than	2860 (6,300)	3270 (7,200)	4080 (9,000)	3450 (7,600)	3860 (8,500)	3270 (7,200)	3450 (7,600)	3860 (8,500)
c. Centerline of rear axle to rear tow bar mm (inches)	1700+50,-75 (67+2,-3)	1850±75 (73±3)	1850±75 (73±3)	--- ---	--- ---	1850±75 (73±3)	--- ---	--- ---
d. Centerline of bogie to rear tow bar mm (inches)	--- ---	--- ---	--- ---	2490+50,-75 (98+2,-3)	2490+50,-75 (98+2,-3)	--- ---	2490+50,-75 (98+2,-3)	2490+50,-75 (98+2,-3)
e. Wheelbase mm (inches) not less than/ not more than	4320 (170)	4770 (188)	4770 (188)	4700/5310 (185/209)	4700/5310 (185/209)	4770 (188)	4700/5310 (185/209)	4700/5310 (185/209)
f. Boom capacities, each boom kg (tons) minimum	7260 (8)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)
g. Snatch blocks, each, capacity kg (tons), not less than	7260 (8)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)	10 880 (12)
h. Winch rating: Each winch kg (pounds) not less than	7260 (16,000)	10 880 (24,000)	10 880 (24,000)	10 880 (24,000)	10 880 (24,000)	10 880 (24,000)	10 880 (24,000)	10 880 (24,000)
i. Winch cable class	6x19	6x19	6x19	6x19	6x19	6x19	6x19	6x19
j. Cable size mm (inches), not less than	14 (9/16)	16 (5/8)	16 (5/8)	16 (5/8)	16 (5/8)	16 (5/8)	16 (5/8)	16 (5/8)
k. Cable length m (feet), not less than	46 (150)	46 (150)	46 (150)	46 (150)	46 (150)	46 (150)	46 (150)	46 (150)
l. Cable type	Preformed, extra improved plow steel cable with steel core.							

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3.6.1 Type II wrecker unit. The wrecker unit shall be a welded steel topping lift frame, mounting swinging booms on each side, incorporating stabilizing outriggers, two power operated load line winches, and two hand operated topping lift winches, as a unit assembly.

3.6.2 Type II body structure. The body structure shall conform to 3.5.2.

3.6.3 Type II front push bumper. A front push bumper as specified in 3.5.3 shall be provided.

3.6.4 Type II backplate. The backplate shall be constructed of minimum 6.4 mm (0.25 inch) steel plate, flanged at the bottom, reinforced and mounted to transfer the loads imposed on the back chassis frame. The backplate shall be the full width of the vehicle and not less than 560 mm (22 inches) from the bottom flange to the top edge of the plate. Slots for scotch block chains shall be furnished. Taillights, stoplights, and directional signal lights shall be inset into the backplate.

3.6.5 Type II frame structure. A welded steel topping lift frame shall be provided for the two swinging booms and shall be supported with a back stay. The frame structure shall support the load and the topping lift winches as well as the booms and outriggers; shall be mounted forward of the rear spring front hanger bracket; and shall be attached to and supported by the truck frame.

3.6.6 Type II load line winches. The winch drum pull rated capacity with a single line on the bare drum shall have a safety factor of not less than 2.0. Winches shall conform to or exceed the requirements of SAE J706. The power takeoff, operable from the cab, shall be of a capacity to operate the winches simultaneously. Separate and independent forward and reverse winch drum drives shall include machine-cut gears, clutch and self-locking worm and wheel. Worm gearing shall be alined, lubricated, and housed in a dust- and oil-tight case. A means shall be provided for free unreeling of the load lines. An automatic drum brake shall prevent overtravel of drum when free-unreeling. Load lines shall be equipped with hooks of sufficient capacity for rated loads.

3.6.7 Type II winch and throttle controls. Two sets of winch and throttle controls shall be provided, one on each side at the rear. Throttle control head shall be corrosion-resistant twist lock, push-button lock, or micro-adjustable mechanism. Push-pull cable control shall be recessed. Control cables shall be stainless steel with a neoprene or equivalent outer cover. Control cables or rods shall be routed and anchored to prevent damage and interference with operations by snow and ice. Operating positions of the controls shall be shown on corrosion-resistant metal or weather-resistant plastic instruction plates. Corrosion-resistant metal or weather-resistant plastic

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plates inscribed with the payload capacity in the lift-tow load position shall be provided, one near each set of controls. When space is not available in the sides of the vehicle at the rear to permit recessing the two sets of winch and throttle controls, two sets of upright type controls may be furnished.

3.6.8 Type II topping lift and winch. The topping lift shall be reeved for the specified boom capacity and shall be provided with a pull line running over a fairlead sheave to a hand-crank operated winch.

3.6.9 Type II booms. Each boom shall traverse not less than 120 degrees from the locked-together position at the rear. Each boom shall be supplied with a top sheave having cable guides and hook loop. The load line shall pass over a swivel fairlead sheave with a cable guide on the topping lift frame, to the power winch drum. The load line shall be directly operable from the mast swivel sheave for recovering heavy loads.

3.6.10 Type II outriggers. Manual outriggers shall be attached to the frame structure, one on each side, and shall be adjustable for varying ground distances. Outriggers shall be hinged near the top of the frame structure to swing outward to not less than 35 degrees. Each leg of the outrigger shall be telescoping, shall have a ground pad not less than 410 cm^2 (64 square inches) in area, and shall be equipped with bright zinc steel stay chains. When specified (see 6.2), rear ground jacks shall be furnished.

3.6.11 Type II lift-tow load. Without the use of counterweights, the wrecker shall have sufficient front end curb weight to provide the lift-tow load capacity specified in line "b," table XIII, with not less than one half of the front axle curb weight maintained on the front axle. The actual lift-tow load capacity shall be determined as follows: half the front axle curb weight of the wrecker multiplied by the wheelbase of the wrecker and divided by the distance to the lift-tow load position (as defined in table XIII). The wrecker shall be capable of lifting the front of the wrecker off the ground by use of a load in the lift-tow load position, without any damage to the truck wrecker.

3.6.12 Type II accessories. The following accessories shall be provided:

(a) Two snatch blocks with hooks for double reeving
(b) Universal steering wheel clamp
(c) Telescoping tow hitch, with face bar attached, with rated capacity not less than the wrecker GVW. Two bright zinc steel tow chains of high-test coil grade with J-hook on one end and T-hook and grab hook on the other end, 16 mm by 2440 mm (5/8 inch by eight feet) in length, conforming to SAE J1142. Crossbeam and spacer blocks in accordance with SAE J1142 shall also be provided
(d) Two scotch blocks with chains

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(f) One crank for topping lift winch
(g) Two 13 mm (1/2-inch) (for size 16T wreckers) or two 16 m (5/8-inch) (for all other size wreckers), high-test, bright zinc steel safety chains, each 3660 mm (12 feet) long, with a clevis at one end for attaching the chain to an eye on the boom; with a grab hook with spring loaded safety latch on the other end for attaching the chain to a vehicle being towed
(h) Three emergency reflective triangles conforming to Federal Motor Carrier Safety Regulation 393.95(f)(2)(i)
(i) A steel stowage box or compartment conforming to 3.5.13(g)
(j) A towing sling conforming to 3.5.13(h).

* 3.6.13 Type II electrical accessories. The following electrical accessories shall be installed:

(a) Two swivel type, sealed-beam floodlights as specified in 3.5.14(a)
(b) Undercarriage mounted work light(s) as specified in 3.5.14(b)
(c) In addition to the brake lights specified in 3.1.1.15, two elevated brake lights actuated upon application of the service brakes. Brake lights shall be suspended from the mast and shall be located approximately 1270 mm (50 inches) on center
(d) An amber, rotating, vehicular warning light conforming to MS 51317-4. The rotating light shall be mounted with reinforcement at point of attachment on vehicle centerline for 360-degree visibility. The warning light shall be operated by means of a separate switch with an indicator light located on the instrument panel and marked to indicate the function of the switch
(e) Auxiliary safety light kit conforming to 3.5.14(d)
(f) Wiring conforming to Federal Motor Carrier Safety Regulations 393.27 and 393.28.

3.6.14 Rear-mounted winch. When specified (see 6.2), a winch shall be located at the center of the body to the rear of the boom masts. The winch shall be mechanical or hydraulic motor driven. The winch mountings shall be capable of withstanding all loads imposed. The winch shall conform to SAE J706 and shall have a single line pull of not less than 89 kN (20,000 pounds) on a bare drum. The winch shall be wound with not less than 61 m of 16 mm (200 feet of 5/8-inch) diameter, preformed, 6x37, improved plow steel, independent wire rope core (IWRC) and shall be equipped with an end chain and hook. A roller guide shall be furnished to the rear of the winch to control pay in and pay out of wire rope over rear of wrecker body. Controls shall be located as specified in 3.6.7.

3.7 Workmanship. Defective components or parts and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished. Welded, bolted and riveted construction utilized shall be in accordance with the highest standards of the industry. Vehicles shall have a quality appearance. Doors, hoods and compartment openings shall not have excessive gaps in the fit of the item. Paint shall be uniform without runs or sags.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection as part of manufacturing operations is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Government verification. Quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of observation of the operations to determine that practices, methods, and procedures of the contractor's inspection are being properly applied. Failure of the contractor to promptly correct product deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to specification criteria has been demonstrated.

4.3 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. The purpose of the inspection shall be to determine vehicle conformance to the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its rights under the provisions of the contract.

* 4.3.1 Road test. The first production vehicle shall be examined and road tested by the contractor. The vehicle, loaded with its required lift-tow load and all the equipment specified herein, shall be driven at speeds up to at least 80 km/h (50 mph) for a minimum distance of 32 kilometers (km) (20 miles). In addition, wreckers with wheel-lifts shall be road tested for a distance of not less than 8 km (5 miles) towing a vehicle equal to the maximum rated size for lift-tow load for that wrecker and for 8 km (5 miles) towing a subcompact size automobile. The distance traveled shall consist of highway, rough dirt roads, partly surfaced gravel roads, puddles

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and sharp turns. For each 8 km (5 miles) of travel, at least one sudden stop shall be accomplished. The vehicle shall be observed for handling characteristics consistent with safe operating conditions. Upon completion of the road test all equipment shall be thoroughly examined. The equipment shall show no evidence of damage; all doors shall be opened and shut; there shall be no evidence of mud or dust inside; and there shall be no evidence of racking of the body.

4.3.1.1 Front wheel drive conversion units. In addition to the road test in 4.3.1, the vehicle, for all-wheel drive models qualifying under 3.1.1.16, shall be road tested loaded to gross vehicle weight rating (GVWR) and without payload. Each axle shall be loaded to its gross axle weight rating (GAWR) for the test. If this cannot be accomplished with the wrecker body installed, it is acceptable for an alternate method of loading the chassis with a payload to establish gross axle weight rating such as temporarily mounting a stake body for carrying the payload. This test shall be performed and chassis accepted prior to inspection and testing of the completed vehicle as a wrecker unit. The road test for loaded and unloaded conditions shall be for not less than 48 km (30 miles) each at speeds up to 80 km/h (50 mph) over highways and gravel roads and for not less than 8 km (5 miles) each at speeds up to 24 km/h (15 mph) over cross country terrain with ground and grade requiring all wheel drive. During the loaded and empty portions of the road test, the brakes shall be applied firmly, bringing the vehicle to a sudden stop no less than 5 times during each portion of the road test. During the road test, the front and rear suspension and the drivetrain shall be periodically inspected for interference and contact with other vehicle components. Abnormal contact of the drivetrain or suspension components with other components shall be cause for rejection. Front spring bumpers shall not make contact with frame stops except under the most extreme cross country conditions. Operational or mechanical failures of vehicle components during the road test shall be cause for rejection. Failure includes permanent deformation as well as breakage.

4.3.2 Service brakes. With the vehicle loaded with its required lift-tow load, the vehicle shall be tested to determine conformance to Federal Motor Carrier Safety Regulation 393.52.

4.3.3 Hydraulic winch and boom test. With the outriggers in place where required and the boom extended to position “j” in table X, type I hydraulic wrecker shall lift a weight equal to the extended rating specified in line “h” of table X. The test weight shall be raised to a height at least equal to the height of the deck of the wrecker. With the weight suspended at the height of the wrecker deck, the height to the tip of the boom from the ground shall be measured and the power shall be shut off. The weight shall remain suspended for a minimum of 30 minutes and the tip of the boom shall descend not more than 13 mm (1/2 inch) during the test period. Following the capacity weight test, the wrecker shall be tested to verify conformance to 3.5.1.2, 3.5.1.4, and,

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when applicable, 3.5.12. When the tests are completed, the entire unit shall be inspected for evidence of cracks or deformation. With the boom in the lift-tow position, the hooks shall be attached to a weight or fixture so that when the winch is engaged the front of the wrecker will lift off the ground. Following this test the wrecker and chassis shall be examined for evidence of cracks or distortion.

4.3.4 Winch capacity verification. The contractor shall certify test results showing compliance with the winch capacity specified in table X or table XIII, with the safety factor specified in 3.5.5 or 3.6.6.

4.3.5 Wrecker lift-tow load capacity verification. The contractor shall perform calculations verifying conformance to the lift-tow load in accordance with 3.5.11 or 3.6.11. The front axle curb weight (load imposed at the ground by the front axle of the vehicle) that is used in the calculations shall be verified by weighing in the presence of the Government representatives.

* 4.3.6 Rear axle weight. The wrecker, loaded to its most severe lift-tow load configuration in accordance with tables X, XII or XIII, as applicable, shall have the load on the rear axles weighed. The weight measured shall not exceed the rear GAWR.

4.3.7 Production sample. Upon acceptance of the first production vehicle, it shall remain at the manufacturing facility as a production sample, and shall be the last vehicle shipped on the contract. The contractor shall maintain the vehicle in an as new condition for the duration of the contract.

4.3.8 Failure. Failure of the first production vehicle to meet requirements of the contract shall be cause for the Government to refuse acceptance of all vehicles under contract until corrective action has been taken.

4.4 Inspection of production vehicles. The contractor's inspection system shall as a minimum assure that the vehicle conforms to the physical and dimensional requirements and is capable of meeting performance requirements contained herein. For each vehicle under contract, the contractor shall make available to the Government, at the point of final acceptance, records acceptable to the Government indicating that the servicing and adjusting required by 3.4.28 have been accomplished.

4.5 Acceptance inspection. Each completed production vehicle shall be operated for a distance of not less than 8 km (5 miles) at speeds up to and including 80 km/h (50 mph). During the acceptance test the vehicle shall be observed for handling characteristics consistent with safe operating conditions. Following the 8 km (5 mile) road test the boom on type I shall be raised to at least 45 degrees, lowered, fully extended and retracted to the normal travel position without

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the use of weights. Following these tests the vehicle shall be inspected for hydraulic leaks. Any evidence of leakage that results in a droplet shall be considered a failure. The winch and boom controls shall operate smoothly and shall return to a neutral hold position when released.

4.5.1 Failure. Failure of any vehicle to pass the acceptance test shall be cause for the Government to refuse acceptance of the vehicle until corrective action satisfactory to the Government has been taken.

5. PACKAGING

5.1 Vehicle processing. The vehicle shall be processed for shipment, from the manufacturer's plant to the initial receiving activity, in accordance with the manufacturer's standard commercial practice.

6. NOTES

6.1 Intended use. The wreckers covered by this specification are intended for nontactical use by the Government for recovering, lifting and towing disabled vehicles or equipment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, class, size, and model of wrecker required (see 1.2).
(c) Issue of DODISS to be cited in the solicitation and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Identification of appropriate military service for painting (see 3.1.1.1).
(e) Identification of appropriate military service for marking (see 3.1.1.2).
(f) Concealed markings, if required (see 3.1.1.2).
(g) Rustproofing, if required (see 3.1.1.3).
(h) Tropical rustproofing, if required (see 3.1.1.3).
(i) Trailer lighting cable, if required (see 3.1.1.9).
(j) Electric trailer brake controller, if required (see 3.1.1.10).
(k) Silicone brake fluid, if required for class 10 (see 3.1.1.11).
* (l) An auxiliary 120 volt generator unit, if required (see 3.1.1.17).
(m) Satisfactory performance on JP-4, JP-5 and JP-8 fuels, if required (see 3.4.1.1).
(n) Gasoline engine, if required (see 3.4.1.2).
(o) Silicone rubber hoses, except for class 10, if required (see 3.4.1.8).
(p) Power plant heaters and fuel warmer, if required (see 3.4.1.9).
* (q) Fuel fired engine preheater, if required (see 3.4.1.10).
* (r) Vernier throttle control, if required (see 3.4.1.11).

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* (s) Starting motor circuit breaker, if required (see 3.4.2.1).
(t) Alternator capacity, if other than as specified (see 3.4.2.3).
(u) Auxiliary 24-volt system with trailer receptacle, if required (see 3.4.2.7).
* (v) Fuel and water separator, if required (see 3.4.3.3).
* (w) A spark arresting muffler. if required (see 3.4.4.1).
(x) Automatic transmission, if required (see 3.4.5.3).
(y) Two-speed transfer case for models 4x4 and 6x6, if required (see 3.4.5.4).
(z) Two-speed axle for models 4x2 and 6x4, if required (see 3.4.9.2).
(aa) Traction control, if required (see 3.4.9.3).
(ab) Wide base tires and wheels, if required (see 3.4.10).
(ac) Disc type wheels, if required (see 3.4.10).
(ad) Bias ply tires, if required (see 3.4.10.1).
* (ae) Low profile tires, if required (see 3.4.10.1).
(af) Spare wheel or rim, if required (see 3.4.10.4).
(ag) Spare tire assembly, if required (see 3.4.10.5).
(ah) Brake controls for use from a second towing vehicle, if required (see 3.4.11.4).
* (ai) Increased braking capability, if required (see 3.4.11.5).
(aj) Tilting hood and fender assembly, if required (see 3.4.12).
(ak) Tilt type cab, if required (see 3.4.12).
* (al) Intermittent windshield wipers, if required (see 3.4.14).
(am) Tools, if required (see 3.4.17.1).
* (an) Remote control curbside rearview mirror, if required (see 3.4.21).
* (ao) Heated rearview mirrors, if required (see 3.4.21).
(ap) Air operated horn, in addition, if required (see 3.4.22).
(aq) Air conditioning, if required (see 3.4.25).
* (ar) Front mounted winch, if required (see 3.4.26).
* (as) AM/FM radio, if required (see 3.4.27).
* (at) Cooling system protection down to -54°C (-65°F), if required (see 3.4.28.1).
(au) Extended boom for sizes 40T and 45T, if required (see 3.5.1).
(av) If rear spades are not required on type I (see 3.5.10).
(aw) Wheel-lift towing device, type I hydraulic wrecker only, if required for size 25T (see 3.5.12).
(ax) Automobile towing sling, if required for sizes 35T, 40T and 45T (see 3.5.13(h).
(ay) Hydraulic fluid type MIL-H-83282, if required (see 3.5.15.3).
(az) Rear mounted winch, if required (see 3.6.14).

6.3 Performance prediction. SAE Truck Ability Prediction Procedure computations and computations for low speed and maximum geared speed will be required by the contract. The SAE Work Sheet Item 1 should include vehicle model number, engine model number, and vehicle type, class, size and model. Unless other conditions are cited in the contract, computations should be made for normal atmospheric pressure, normal ambient air temperature, and still, dry

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air. The factors to be used in predicting truck ability (see 3.3.1.1) are established as follows for the corresponding SAE Truck Ability Prediction Tables:

Table 1	- Tire Factor. This factor must relate to the size of tires furnished by the contractor in accordance with this specification.
Table 2	- Altitude Factor. 1.00
Table 3	- Rolling Factor. 1.613
Table 4	- Area Factor. 0.173
Table 5	- Velocity Factor. 250.0
Table 6	- Altitude Factor. 1.00
Table 7	- Chassis Friction Horsepower. Use applicable power unit GVW (to nearest, higher, 454 kg (1,000 pounds)) and the engine rpm (to nearest 100 revolutions) which is required for 80 km/h (50 mph) geared speed. For GVW and engine speed beyond the range of this table, factors shall be extrapolated.
Table 8	- Grade Factor. 0.75
Table 8A	- Correction Factor. Not required.
Table 9	- Road Factor. 0.0.

6.4 Change in classification. The initial issue of this document covered only hydraulic wreckers, as follows:

Previous (initial) class	Current class, type I
A	10
B	28
C	32
D	39.5
E	50

Mechanical wreckers were formerly covered by specifications KKK-T-2107 through 2111, as follows:

Previous coverage, type V in	Coverage in this document:
KKK-T-2107, class D	Type II, class 21, model 4x2
KKK-T-2108, class A	Type II, class 24, model 4x2
KKK-T-2108, class C	Type II, class 32, model 4x2
KKK-T-2109, class A	Type II, class 34.5, model 6x4
KKK-T-2109, class C	Type II, class 44.5, model 6x4
KKK-T-2110, class C	Type II, class 24, model 4x4
KKK-T-2111, class A	Type II, class 36, model 6x4
KKK-T-2111, class B	Type II, class 44, model 6x6

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6.5 Subject term (key word) listing.

Commercial vehicle
Elevating boom
Hydraulics
Lift-tow load
Nontactical vehicle
Spades
Swinging booms
Telescopic boom
Truck, commercial
Truck, nontactical
Truck, wrecker
Wheel-lift
Winches
Wrecker, hydraulic.

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content regardless of the marginal notations and relationship to the previous issue.

Custodians:
Army - AT
Navy - YD
Air Force - 99

Preparing activity:
Army - AT

Project No. 2320-0576

Review activities:
Air Force - 84

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